COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit 3 : Session 6 : SLO 1 — Array Multiplier and Wallace Tree Multiplier
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1. Label the Diagram (Remembering) — 2×2 array multiplier

• Inputs: A0, A1, B0, B1
• Four AND gates produce the partial-product bits A0B0, A1B0, A0B1, A1B1.
• One Half Adder and one Full Adder combine the partial products.
• Outputs: Product bits P0, P1, P2, P3.

2. Match the Concepts (Understanding)

• 1. Array Multiplier → b. Uses a systematic arrangement of adders to sum partial products
• 2. Wallace Tree Multiplier → c. Uses a tree-like structure of adders to sum partial products in parallel
• 3. Booth's Algorithm → a. Reduces the number of partial products by grouping multiplier bits

3. Step-by-Step Process (Applying): A = 11 (3) × B = 10 (2)

• Step 1 — partial products: PP1 = A × B0 = 11 × 0 = 00; PP2 = A × B1 = 11 × 1 = 11
• Step 2 — add the shifted partial products: 00 + 110 (PP2 shifted left by one) = 0110
• Final binary product P3P2P1P0 = 0110 = 6 in decimal (3 × 2 = 6 ✓)

4. Compare and Contrast (Analyzing)

• Structure — Array: regular and systematic | Wallace Tree: irregular tree-like structure
• Speed (delay) — Array: slower, delay depends on array depth | Wallace Tree: faster, logarithmic delay due to parallel summation
• Hardware complexity — Array: simpler and more regular | Wallace Tree: more complex and irregular
• Primary advantage — Array: simple, regular layout, easy to design and expand | Wallace Tree: higher speed

5. Design a Concept (Creating)

• Choice: the Wallace Tree Multiplier.
• Reasoning: the requirement is the fastest possible multiplier for real-time signal processing. The Wallace tree sums partial products in parallel using a tree of carry-save adders, which reduces the delay dramatically compared with the sequential ripple additions of an array multiplier. Although more complex, its superior speed and low latency make it ideal for performance-critical applications.